

Saumya Parekh

Civil Engineering Undergraduate | Computational Civil Engineering

Ahmedabad, Gujarat | Mail | LinkedIn
GitHub | Portfolio-website

PROFESSIONAL SUMMARY

Civil Engineering undergraduate specializing in the intersection of traditional infrastructure and modern technology. Proven experience developing engineering software, IoT-based laboratory telemetry systems, and AI-driven platforms. Passionate about applying computational methods, automation, and smart sensing to solve complex geotechnical and construction challenges.

EDUCATION

Pandit Deendayal Energy University (PDEU)
B.Tech. in Civil Engineering

2024 – 2028
Ahmedabad, Gujarat

TECHNICAL SKILLS

Engineering Domain: Concrete Technology & Mix Design (IS 10262:2019), Geotechnical Monitoring, Surveying, Construction Materials.

Software & GIS: AutoCAD, QGIS, Pix4D.

Programming & Tech: Python, Flask, OpenCV, C/C++ (Embedded), IoT (ESP32-CAM), Computer Vision.

TECHNICAL PROJECTS & HACKATHONS

PRO-MIX – Automated Concrete Mix Design Platform

Independent Project

- Developed an automated platform for IS 10262:2019 concrete mix design featuring engineering validation and automated report generation.
- Engineered supplementary cementitious material comparison and geopolymers analysis modules.

Universal Intelligent Compaction (UIC) Kit

NHAI Innovation Hackathon

- Architected a research concept proposing an intelligent retrofit kit for conventional road rollers.
- Designed digital construction workflows utilizing sensor fusion and edge computing to modernize compaction methods.

Automated Soil Strain & Settlement Monitoring System

Team Project

- Built a laboratory telemetry platform integrating ESP32-CAM, strain gauges, and ultrasonic sensing.
- Utilized Python, Flask, and OpenCV to enable real-time monitoring and data visualization of soil strain, settlement, and load behavior.

Smart Crop & Scheme Advisor

IIT Gandhinagar AgriTech Hackathon

- Co-developed an AI-powered agricultural decision-support platform integrating IoT, machine learning, and weather intelligence.

- Implemented multilingual assistance and government scheme recommendation systems to aid financial forecasting for farmers.

EXPERIENCE & CAMPUS INVOLVEMENT

CSSI Intern

Sabar Seva Trust

May 2025 – Jun 2025

- Contributed to community development initiatives and field activities, leveraging collaborative problem-solving to address local operational challenges.
- Managed field documentation and project reporting.

Content Writer & Creative Team

Brahmand Club

2025

- Created technical and promotional copy for university events, driving engagement for digital outreach campaigns.

CONFERENCES & PROFESSIONAL DEVELOPMENT

National Seminar on Ground Engineering Problems: Nirma University

Apr 2026

World of Concrete India Expo: Mumbai

Oct 2025

Advancement in Geotechnical Engineering for Deep Excavations: CEPT University

Aug 2025